# BEHAVIORAL FINANCE

FINANCIAL ECONOMICS – SPRING 2026

# RATIONALITY IN ECONOMICS

- Thaler (2016), referring to a “psychologically realistic depiction of the representative agent” that does not substitute the neoclassical rationality paradigm, which should be augmented “with additional descriptive theories that are derived from data rather than axioms”

- Simon (1955) suggested that to make economic models more useful “the task is to replace the global rationality of economic man with a kind of rational behavior that is compatible with the access to information and the computational capacities that are actually possessed by organisms, including man, in the kind of environments in which such organisms exist”

# QUESTION

- Question: which experiment do you choose?

Experiment 1: (A) a 61% chance of winning €520,000, (B) a 63% chance of winning €500,000?

Experiment 2: (A) a 98% chance of winning €520,000, (B) a 100% chance of winning €500,000?

# ANSWER

- Experiments 1 and 2 in section 5.2 describe the "Allais paradox"
- In most cases people choose A in experiment 1 and B in 2, even if this choice violates the principle of expected utility. In fact, Experiment 2 differs from Experiment 1 because in both Choices A and B there is a 37% chance of winning. Therefore, if A is preferred to B in Experiment 1, the same choice should be made in Experiment 2.
- The interpretation is that people overweight outcomes that are certain giving rise to a “certainty effect”. If the monetary sums in the previous experiments are negative rather than positive (for example, losing €500,000 rather than gaining €500,000) preferences switch and prevails loss aversion.

# BEHAVIORAL FINANCE

- Kahneman (2011) argues that the rational model is so hard to implement, requiring full information and full understanding of the consequences of all possible choices, that “reasonable people cannot be rational by that definition”.

- Bob Shiller and Dick Thaler, Nobel Prize winners in 2013 and 2017, were among the architects of the movement at the basis of the development of "behavioral finance", which borrowed many concepts from psychology thanks to scholars such as Kahneman who worked on the border of the two disciplines.

# BEHAVIORAL FINANCE THEORY: THINKING FAST/SLOW

- Kahneman (2011) proposes a description of individuals' decisions as the result of the interaction between two systems, the fast system (system 1) and the slow system (system 2):
- system 1 operates quickly and automatically, using the association of memories and images, prefers simple and reassuring explanations, even if obtained with few data. This leads individuals to dwell on the first impression and believe that the available evidence is sufficient to deal with a certain situation.
- system 2, allocating mental resources to the tasks to be done at a certain time and requiring attention and concentration.

# BEHAVIORAL FINANCE THEORY: THINKING FAST/SLOW

- Typically, the two systems work together, and System 2 is only activated when System 1 cannot handle a situation. One of System 2's functions is to monitor what System 1 suggests and interfere when it thinks actions are wrong, but only if the stakes are high enough.

- Sometimes System 1 deceives System 2 by creating illusions. System 1 performance can improve through experience that make someone become an “expert” in a field with the associated ability to quickly decide, typically in the right way, on a new case.

# BEHAVIORAL FINANCE THEORY: THINKING FAST/SLOW

- System 1 often uses the causality hypothesis. If there was adequate statistical knowledge, System 2 could intervene to correct the interpretations given by System 1, even though there may be cases when System 2 is too lazy and/or the cost of error is low.

# BEHAVIORAL FINANCE THEORY: HEURISTICS

- Heuristics are a reasoning scheme that is based on association: if a problem is too difficult to solve, a simpler one is solved by trying to find an answer that also applies to the first one. Therefore, heuristics is a rule of thumb used by individuals for making decisions in a complex and uncertain environment, especially when time is limited.

- Representativeness
- Availability
- Anchoring

# TEST

- Which event do you think is most likely next year:

(A) a flood in North America that will kill more than 1000 people,

(B) an earthquake in California that will cause a flood in which more than 1000 people will die.

# BEHAVIORAL FINANCE THEORY: REPRESENTATIVENESS

- Tendency to calculate the probability of an event based on its superficial similarity to a phenomenon rather than on objective probabilities. If, for example, one must guess the profession of an individual who is described as shy and quiet and with a predisposition to help others, it is more likely that, in the absence of any other information, the individual is supposed to work in a library instead of being a lawyer. There is a tendency to use information about the character of the individual, ignoring the evidence that there are many more lawyers than library clerks in the United States.

# BEHAVIORAL FINANCE THEORY: AVAILABILITY

- Availability: tendency to calculate the probability of an event based on the ease with which examples of that event can be remembered, for example, the probability of a heart attack in a middle-aged person is estimated based on the examples that occurred among one's acquaintances.

# BEHAVIORAL FINANCE THEORY: ANCHORING

- The tendency to calculate the probability of an event from a prior that acts as a reference anchor, without using available information to update the probability in a rational way. The prior that serves as a reference point may or may not be objectively relevant to the evaluation . In one experiment, researchers ask what percentage of countries adhering to the United Nations are after spinning a "wheel of fortune" that in theory should stop on one of the hundred numbers between 1 and 100 but in reality is rigged to give only 10 or 65.

- The average value declared by the sample receiving the signal '10' is 25% and the average value declared by the sample receiving '65' is 45%

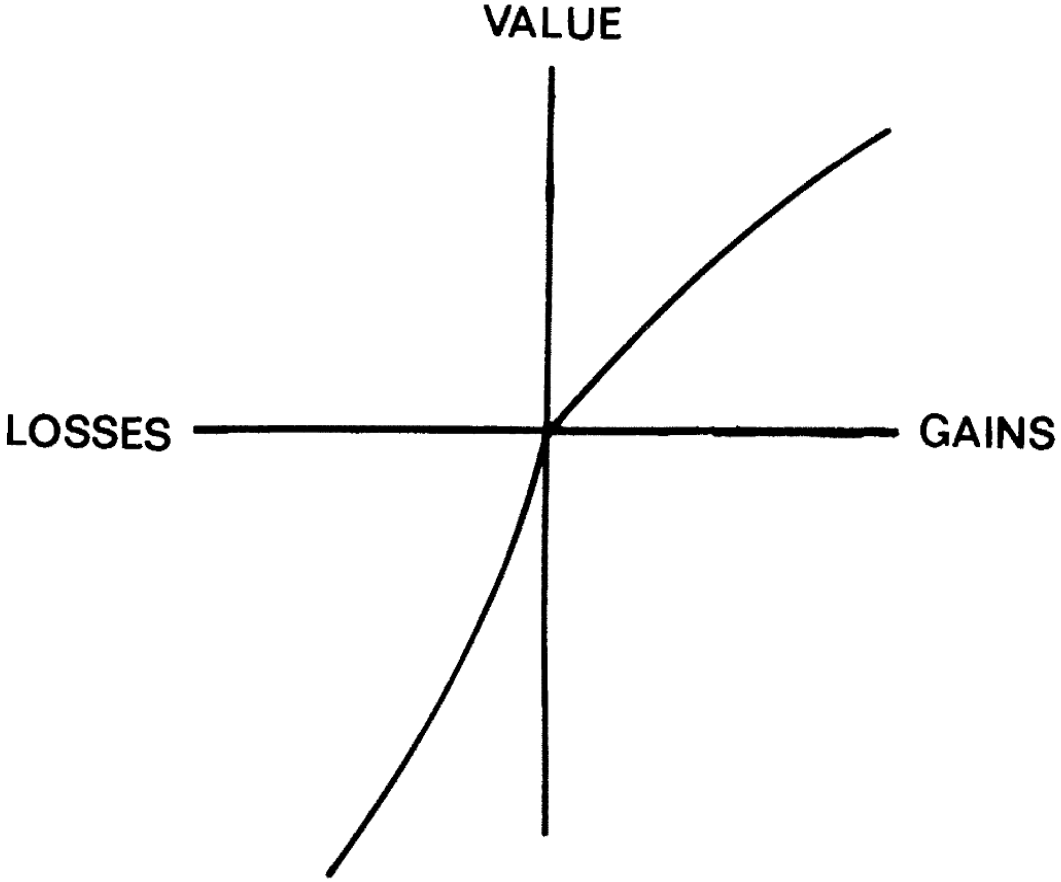
# BEHAVIORAL FINANCE THEORY: MENTAL ACCOUNTING

- Breaking complex financial decision-making into smaller, more manageable parts, examples:
  - wealth in different accounts
  - gambling and house money effect and the “roaring 20s”
  - evaluating asset classes differently

# BEHAVIORAL FINANCE THEORY: PROSPECT THEORY

- This theory suggests that investors are risk averse when they gain (therefore they need a higher risk premium and are more likely to sell if the expected return is not large enough) and risk lovers when they lose (so they are likely to hold for a low expected return)

- This may explain the disposition effect, investors holding losers and selling winners



# BEHAVIORAL FINANCE THEORY: PROSPECT THEORY

- This may explain the disposition effect, investors holding losers and selling winners

- Barber and Odean (2013) document that the probability of selling stocks up between 18-22% since purchased is 2.65 (1.77) times greater than the probability of selling stocks that have experienced returns between -2% and 2% (-18%-22%).

- difference between individual and institutional investors

- buy low/sell high

# BEHAVIORAL FINANCE THEORY: ENDOWMENT EFFECT

- Suppose the original price of a ticket for a concert was equal to €200 (primary market). After a month the secondary market price is €400. If you bought the ticket in the primary market, then the current situation is described by the presence of the ticket for a concert, and the value function describes changes in value relatively to the current situation. You are unlikely to sell the ticket to someone else, regardless of the price increase, due to the steep slope of the value function when you move to the left of the origin.

# BEHAVIORAL FINANCE THEORY: ENDOWMENT EFFECT

- If you did not buy the ticket in the primary market, and therefore if the current situation is described by the absence of a ticket for a concert, then you are less likely to spend money to buy the ticket because the value function is less steep to the right of the origin. The endowment effect we overvalue what we have (the ticket in one case, money in the other) and we are not willing to face a large loss by changing the current situation.

# BEHAVIORAL BIASES

- Our mind is not built to use statistical analysis, rather it relies on intuition…we overlie on small samples

- Do we try to learn what we do not know or confirm what we know already? Google is on my side

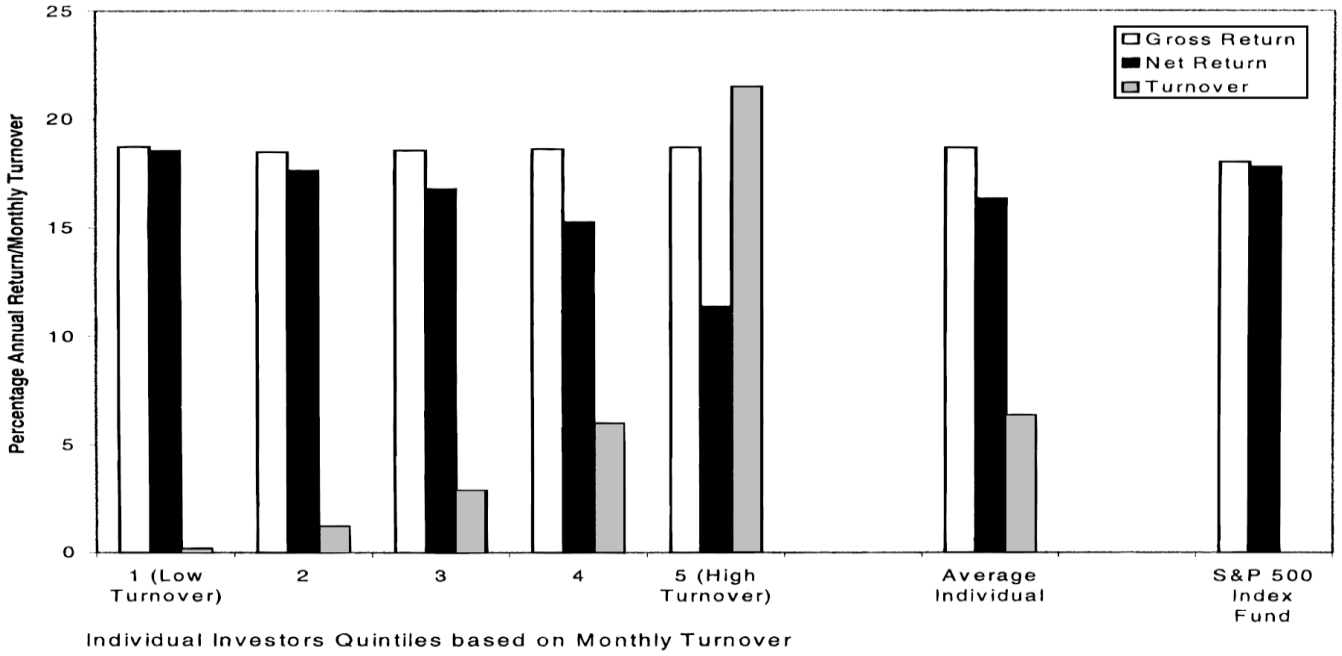
- Anchoring

# EMOTIONAL BIASES

- Optimism and overconfidence
- Self control problems
- Regret aversion
- Framing biases

# FINANCIAL ERRORS

- No risk taking
- Overtrading
- Lack of diversification



# SOLUTIONS

- Policies, particularly zero cost policies

- Financial education